Agent-Based Monte Carlo Simulation of Tumor Growth

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1 Introduction

In silico simulations are increasingly used to study tumor evolution, growth, and progression, as they allow one to explore complex biological processes in a controlled and systematic way. Such approaches are particularly useful when spatial effects, local interactions, and stochasticity play an important role, as is the case in solid tumor growth. In this project, tumor growth is modeled using an agent-based Monte Carlo approach. Agent-based models (ABMs) describe a system in terms of individual cells that follow simple, local rules depending on their state and immediate surroundings. In cancer growth modeling, ABMs are well suited to capture spatial heterogeneity and emergent behavior arising from cell–cell interactions and environmental constraints. A common formulation employs on-lattice models, where cells occupy sites on a regular grid and interact with neighboring cells.

The stochastic nature of cellular processes such as division, death, and mutation is naturally incorporated using Monte Carlo methods, in which events occur probabilistically at each time step. The model is implemented in the Julia programming language and deliberately relies on simple data structures and basic random number generation, without the use of specialized or high-level simulation packages.

The aim of this work is to qualitatively study tumor growth under oxygen-limited conditions using a simple agent-based framework, rather than to provide quantitative predictions. The model is minimal, but can be extended to include additional factors such as nutrients or drug effects. The model assumptions and update rules are described in the following section.

2 Model and Methods

The tumor is simulated on a two-dimensional square lattice of size $N \times N$, where in this work $N = 301$. Each lattice site can be empty or occupied by a normal cell, a cancer cell, or a dead cell. The system is initialized with a single normal cell placed at the center of the lattice. Time evolution proceeds in discrete steps using a Monte Carlo update scheme. At each time step, all living cells are visited once in random order, and stochastic decisions for division or death are made based on local oxygen availability.

A mutation event is introduced by converting a normal cell that has at least one neighboring empty lattice site into a cancer cell.

The mutation time T_{mut} is controlled to explore different growth scenarios, and simulations were performed for $T_{\mathrm{mut}} = 60$ and $T_{\mathrm{mut}} = 120$. Different random seeds were used to generate independent realizations of the stochastic process.

Division and death probabilities depend on both cell type and local oxygen concentration, with the effective parameter values summarized in Table 1. Cancer cells are assigned a higher division probability and a lower death probability compared to normal cells, reflecting their increased proliferative capacity and assumed tolerance to hypoxic conditions. In particular, reduced sensitivity of cancer cells to low oxygen levels is used as a simple way to model hypoxia resistance.

The oxygen field is prescribed as a static, radially symmetric profile that is lowest at the tumor core and increases toward the lattice boundary, mimicking diffusion-limited nutrient supply. It is defined in 1.

$$O(r) = O_{\mathrm{min}} + (1 - O_{\mathrm{min}}) \left(1 - e^{-r/L}\right), \tag{1}$$

where r is the distance from the lattice center, O_{min} denotes the minimum oxygen level, and L controls the characteristic length scale of oxygen recovery.

Cell division is attempted by placing a daughter cell in a neighboring empty site. If no empty site is available, division may still occur through a pushing mechanism, in which a chain of neighboring cells is displaced along a lattice direction until an empty site is reached. Cancer cells are assigned a higher probability to push than normal cells. Dead cells are removed stochastically with a fixed clearance probability, allowing space to be freed over time.

Table 1: Effective probabilities used in the simulations.

Parameter	Normal cells	Cancer cells
Division probability p_{div}	0.30	0.40
Death probability p_{death}	0.05	0.02

The probability for removing dead cells was fixed to $p_{\mathrm{clear}} = 0.02$ for all simulations.

3 Results and Discussion

3.1 Effect of stochastic mutation site

To investigate the influence of stochasticity on tumor evolution, simulations were performed with a fixed mutation time $T_{\mathrm{mut}} = 120$ while varying the random seed. Figure shows representative lattice snapshots for three different seeds at the mutation time (left column) and at a later time $t = 200$ (right column). The mutation location is indicated by a yellow circle in all panels.

Although the mutation time is fixed, the stochastic nature of the model leads to different mutation sites across runs. As a result, cancer cells emerge in regions with different local oxygen concentrations, which strongly influences subsequent growth. For seed 1, the mutation occurs at an oxygen level of $O = 0.671$, and the cancer population remains relatively limited by $t = 200$. In contrast, for seed 4 the mutation arises in a region with higher oxygen availability ($O = 0.716$), leading to rapid proliferation and a pronounced spatial expansion of cancer cells at later times.

Interestingly, seed 6 exhibits a mutation at a lower oxygen level ($O = 0.616$) compared to seed 1, yet shows a stronger cancer expansion by $t = 200$. This indicates that tumor growth is not determined by oxygen availability alone. In this case, local spatial conditions, in particular the availability of neighboring empty sites due to the lattice structure and clearance dynamics, play an important role in shaping the subsequent growth.. These results highlight the combined influence of stochastic mutation placement, oxygen heterogeneity, and spatial constraints on tumor growth within the model.

in the second comparison, the mutation time was varied while keeping the random seed fixed to seed 6 and the time interval to check on the cancer again also fixed to 80 time step. This allows us to explore how the timing of the mutation timing influences growth dynamics, The results are shown in Figure

When the mutation occurs at $T_{\mathrm{mut}} = 90$, cancer cells emerge earlier in the simulation, allowing more time for proliferation and expansion. However, since the mutation arises at a lower oxygen level ($O = 0.616$) compared to the later mutation time, growth is initially limited by hypoxia. By $t = 170$, the cancer population has expanded but remains relatively constrained.

Figure 1: Seed 1 results: left panel shows the mutation point, right panel shows growth at time $t = 200$. O_{mut} is the oxygen concentration at the mutation point.

The second comparison tested seed 6 with a fixed time interval and varied mutation time, to investigate how the mutation time influences the growth behavior of cancer cells.

3.2 Tumor growth dynamics

3.3 Spatial structure

3.4 Effect of oxygen limitation

4 Conclusion and Outlook